

Supplement: Sensitivity of Wireless Chunk Scheduling

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1 Complete Paired Comparison Family

The family includes all full-policy core contrasts and four matched compute-only recovery interventions. All times retain administrative censoring. Intervals are pointwise, while Holm adjustment applies across the complete family. Distinct estimands and adjustment explain why an interval excluding zero may accompany a large adjusted probability.

Cell	Candidate	Comparator	Gain (%)	CI (s)	Raw p	Holm p
churn	Proposed	Compute-retainExpired	4.34	[45.87, 66.33]	0.00010	0.0040
churn	Proposed	Compute-shrink	2.86	[25.53, 48.07]	0.00010	0.0040
churn	Proposed	Proposed-allPrefetch	0.10	[-1.08, 3.55]	0.30307	1.0000
churn	Proposed	Proposed-noCutoff	-0.14	[-4.82, 1.33]	0.28457	1.0000
churn	Proposed	Proposed-noPrefetch	2.86	[25.40, 48.34]	0.00010	0.0040
churn	Proposed	Proposed-noRSSI	0.00	[0.00, 0.00]	1.00000	1.0000
churn	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.00000	1.0000
churn	Proposed	Proposed-retainExpired	1.73	[16.17, 27.10]	0.00010	0.0040
churn	Proposed	Tuned-fixed	-1.72	[-25.66, -15.80]	0.00010	0.0040
combined	Proposed	Compute-retainExpired	30.90	[585.89, 749.59]	0.00010	0.0040
combined	Proposed	Compute-shrink	10.71	[158.43, 199.86]	0.00010	0.0040
combined	Proposed	Proposed-allPrefetch	0.21	[-7.46, 14.33]	0.57044	1.0000
combined	Proposed	Proposed-noCutoff	-0.17	[-5.69, 0.69]	0.14549	1.0000
combined	Proposed	Proposed-noPrefetch	0.85	[1.08, 25.74]	0.05249	0.9974
combined	Proposed	Proposed-noRSSI	10.77	[156.88, 203.95]	0.00010	0.0040
combined	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.00000	1.0000
combined	Proposed	Proposed-retainExpired	6.19	[85.27, 113.14]	0.00010	0.0040
combined	Proposed	Tuned-fixed	-5.25	[-87.00, -62.12]	0.00010	0.0040
uniform	Proposed	Compute-retainExpired	1.51	[13.21, 23.44]	0.00010	0.0040
uniform	Proposed	Compute-shrink	1.51	[13.15, 23.29]	0.00010	0.0040
uniform	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.00000	1.0000
uniform	Proposed	Proposed-noCutoff	0.15	[-1.07, 4.72]	0.23128	1.0000
uniform	Proposed	Proposed-noPrefetch	1.51	[13.07, 23.24]	0.00010	0.0040
uniform	Proposed	Proposed-noRSSI	0.00	[0.00, 0.00]	1.00000	1.0000
uniform	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.00000	1.0000
uniform	Proposed	Proposed-retainExpired	0.00	[0.00, 0.00]	1.00000	1.0000
uniform	Proposed	Tuned-fixed	0.01	[-2.30, 2.48]	0.94691	1.0000
weak_links	Proposed	Compute-retainExpired	25.53	[432.22, 555.91]	0.00010	0.0040
weak_links	Proposed	Compute-shrink	8.42	[113.69, 148.98]	0.00010	0.0040
weak_links	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.00000	1.0000
weak_links	Proposed	Proposed-noCutoff	0.11	[-1.55, 4.91]	0.32787	1.0000
weak_links	Proposed	Proposed-noPrefetch	0.34	[-2.02, 11.57]	0.17648	1.0000
weak_links	Proposed	Proposed-noRSSI	8.75	[121.06, 153.61]	0.00010	0.0040
weak_links	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.00000	1.0000
weak_links	Proposed	Proposed-retainExpired	5.26	[66.69, 93.16]	0.00010	0.0040
weak_links	Proposed	Tuned-fixed	-3.30	[-52.44, -38.97]	0.00010	0.0040
uniform	Compute-shrink	Compute-retainExpired	0.00	[0.00, 0.00]	1.00000	1.0000

Cell	Candidate	Comparator	Gain (%)	CI (s)	Raw p	Holm p
weak_links	Compute-shrink	Compute-retainExpired	18.69	[307.11, 417.79]	0.00010	0.0040
churn	Compute-shrink	Compute-retainExpired	1.52	[14.09, 24.85]	0.00010	0.0040
combined	Compute-shrink	Compute-retainExpired	22.61	[417.38, 560.74]	0.00010	0.0040

2 Completion and Recovery Counters

Data volume is actual transmitted simulator volume. Unaccepted computation includes obsolete and unfinished work; it is not solely duplicate work. Requeued IDs count retry operations, not distinct failed tasks.

Cell	Policy	Complete	Restricted time	Data (MB)	Unaccepted cost	Requeued IDs
churn	Compute-retainExpired	1.00	1280.81	4150.0	80.1	16.5
churn	Compute-shrink	1.00	1261.28	4080.9	1.2	16.5
churn	Proposed	1.00	1225.23	4080.9	1.2	16.5
churn	Proposed-allPrefetch	1.00	1226.48	4081.2	1.2	16.7
churn	Proposed-noCutoff	1.00	1223.54	4080.9	1.2	16.5
churn	Proposed-noPrefetch	1.00	1261.28	4080.9	1.2	16.5
churn	Proposed-noRSSI	1.00	1225.23	4080.9	1.2	16.5
churn	Proposed-noRTT	1.00	1225.23	4080.9	1.2	16.5
churn	Proposed-retainExpired	1.00	1246.81	4150.0	80.1	16.5
churn	Tuned-fixed	1.00	1204.47	3975.5	7.8	2.0
combined	Compute-retainExpired	1.00	2156.53	6070.4	1250.3	131.6
combined	Compute-shrink	1.00	1668.89	4801.0	191.6	100.8
combined	Proposed	1.00	1490.12	4349.7	86.6	40.0
combined	Proposed-allPrefetch	1.00	1493.29	4366.8	89.0	43.0
combined	Proposed-noCutoff	1.00	1487.62	4350.7	86.4	40.4
combined	Proposed-noPrefetch	1.00	1502.92	4327.2	92.3	37.2
combined	Proposed-noRSSI	1.00	1669.92	4865.1	210.7	110.3
combined	Proposed-noRTT	1.00	1490.12	4349.7	86.6	40.0
combined	Proposed-retainExpired	1.00	1588.45	4603.6	363.5	41.6
combined	Tuned-fixed	1.00	1415.82	4069.0	37.0	8.0
uniform	Compute-retainExpired	1.00	1206.24	3966.3	0.0	0.0
uniform	Compute-shrink	1.00	1206.24	3966.3	0.0	0.0
uniform	Proposed	1.00	1188.01	3966.3	0.0	0.0
uniform	Proposed-allPrefetch	1.00	1188.01	3966.3	0.0	0.0
uniform	Proposed-noCutoff	1.00	1189.79	3966.3	0.0	0.0
uniform	Proposed-noPrefetch	1.00	1206.24	3966.3	0.0	0.0
uniform	Proposed-noRSSI	1.00	1188.01	3966.3	0.0	0.0
uniform	Proposed-noRTT	1.00	1188.01	3966.3	0.0	0.0
uniform	Proposed-retainExpired	1.00	1188.01	3966.3	0.0	0.0
uniform	Tuned-fixed	1.00	1188.10	3966.3	0.0	0.0
weak_links	Compute-retainExpired	1.00	1921.57	5473.8	915.2	89.2
weak_links	Compute-shrink	1.00	1562.44	4550.6	152.6	71.2
weak_links	Proposed	1.00	1430.95	4213.9	65.1	24.2
weak_links	Proposed-allPrefetch	1.00	1430.95	4213.9	65.1	24.2
weak_links	Proposed-noCutoff	1.00	1432.59	4220.3	65.6	24.8
weak_links	Proposed-noPrefetch	1.00	1435.78	4182.4	57.8	20.3
weak_links	Proposed-noRSSI	1.00	1568.19	4625.8	142.6	80.2

Cell	Policy	Complete	Restricted time	Data (MB)	Unaccepted cost	Requeued IDs
weak_links	Proposed-noRTT	1.00	1430.95	4213.9	65.1	24.2
weak_links	Proposed-retainExpired	1.00	1510.47	4405.2	267.4	25.6
weak_links	Tuned-fixed	1.00	1385.26	4052.9	24.8	7.0

3 Full Calibration and Evaluation Protocol

3.1 Scenario Settings

Scenario	Parameter	Value
uniform	tasks	240
uniform	workers	8
uniform	heterogeneity	4.0
uniform	rho	0.3
uniform	cost_cv	0.3
uniform	result_ratio	0.1
uniform	dynamic_fraction	0
uniform	weak_fraction	0.0
uniform	uniform_links	True
uniform	slowdown_probability	0.0
uniform	path_loss_exponent	3.0
uniform	reference_dbm	-40.0
uniform	reference_distance_m	1.0
uniform	wall_loss_db	12.0
uniform	shadow_sd	3.0
uniform	horizon	4000
uniform	workload	lognormal
uniform	independent_inputs	False
uniform	mobility	walk
uniform	dip_seconds	20.0
uniform	extra_rtt_ms	0.0
weak_links	tasks	240
weak_links	workers	8
weak_links	heterogeneity	4.0
weak_links	rho	0.3
weak_links	cost_cv	0.3
weak_links	result_ratio	0.1
weak_links	dynamic_fraction	0
weak_links	weak_fraction	0.5
weak_links	uniform_links	True
weak_links	slowdown_probability	0.0
weak_links	path_loss_exponent	3.0
weak_links	reference_dbm	-40.0
weak_links	reference_distance_m	1.0
weak_links	wall_loss_db	12.0
weak_links	shadow_sd	3.0
weak_links	horizon	4000
weak_links	workload	lognormal
weak_links	independent_inputs	False
weak_links	mobility	walk
weak_links	dip_seconds	20.0
weak_links	extra_rtt_ms	0.0

Scenario	Parameter	Value
churn	tasks	240
churn	workers	8
churn	heterogeneity	4.0
churn	rho	0.3
churn	cost_cv	0.3
churn	result_ratio	0.1
churn	dynamic_fraction	0.5
churn	weak_fraction	0.0
churn	uniform_links	True
churn	slowdown_probability	0.0
churn	path_loss_exponent	3.0
churn	reference_dbm	-40.0
churn	reference_distance_m	1.0
churn	wall_loss_db	12.0
churn	shadow_sd	3.0
churn	horizon	4000
churn	workload	lognormal
churn	independent_inputs	False
churn	mobility	walk
churn	dip_seconds	40
churn	extra_rtt_ms	0.0
combined	tasks	240
combined	workers	8
combined	heterogeneity	4.0
combined	rho	0.3
combined	cost_cv	0.3
combined	result_ratio	0.1
combined	dynamic_fraction	0.5
combined	weak_fraction	0.5
combined	uniform_links	True
combined	slowdown_probability	0.5
combined	path_loss_exponent	3.0
combined	reference_dbm	-40.0
combined	reference_distance_m	1.0
combined	wall_loss_db	12.0
combined	shadow_sd	3.0
combined	horizon	4000
combined	workload	lognormal
combined	independent_inputs	False
combined	mobility	walk
combined	dip_seconds	40
combined	extra_rtt_ms	0.0

3.2 Policy Definitions

Policy	Complete switches
Proposed	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=permanent; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Compute-shrink	chunk=shrink; sigma=0.3; fixed=8; link=False; prefetch=none; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Proposed-noRSSI	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=permanent; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=False; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Proposed-noRTT	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=permanent; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=False; compute_weight=True; cancel_expired_transfers=True
Proposed-noPrefetch	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=none; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Proposed-allPrefetch	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=all; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Proposed-noCutoff	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=permanent; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=1.0; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Tuned-fixed	chunk=fixed; sigma=0.5; fixed=1; link=False; prefetch=none; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=True
Proposed-retainExpired	chunk=shrink; sigma=0.3; fixed=8; link=True; prefetch=permanent; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=False

Policy	Complete switches
Compute-retainExpired	chunk=shrink; sigma=0.3; fixed=8; link=False; prefetch=none; recovery=fenced; grace=20; timeout=30; dedup=True; detector=none; hysteresis=False; healthy_probes=3; margin_db=3; hybrid_threshold=0.25; prefetch_cutoff=0.75; speculative=False; rssi_weight=True; rtt_weight=True; compute_weight=True; cancel_expired_transfers=False

3.3 Calibration Knots

Receive power	Goodput (Mbps)	Echo RTT/timeout (ms)	Delivery	Decoded data frames
-130	0	400	0	0
-109	0	400	0	0
-100	0	400	0	0
-93.34	0	400	0	0
-88.06	0	400	0	0
-84.31	0	400	0	0
-81.94	16.24	0.3267	1	2.538e+04
-79.03	21.29	0.3024	1	3.328e+04
-75.28	25.31	0.2863	1	3.956e+04
-70	27.08	0.2862	1	4.231e+04
-60.97	27.08	0.2862	1	4.232e+04
-49.03	27.08	0.2862	1	4.232e+04
-40	27.08	0.2862	1	4.232e+04

Zero decoded frames imply configured receive power, not observed RSSI. Zero echo delivery implies the 400 ms timeout convention, not a measured RTT.

3.4 Fixed-Size Tuning

Candidate	Worst-cell relative regret
Fixed-1	0.000000
Fixed-2	0.053531
Fixed-3	0.099547
Fixed-4	0.129860
Fixed-8	0.270841
Fixed-16	0.515143
Fixed-32	1.099963

Calibration seeds: 4000, 4001, 4002, 4003, 4004, 4005, 4006, 4007, 4008, 4009. Evaluation seeds: 4800, 4801, 4802, 4803, 4804, 4805, 4806, 4807, 4808, 4809, 4810, 4811, 4812, 4813, 4814, 4815, 4816, 4817, 4818, 4819, 4820, 4821, 4822, 4823, 4824, 4825, 4826, 4827, 4828, 4829, 4830, 4831, 4832, 4833, 4834, 4835, 4836, 4837, 4838, 4839, 4840, 4841, 4842, 4843, 4844, 4845, 4846, 4847, 4848, 4849.